

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	Apply advanced methodologies and agile project management techniques to manage software projects effectively		
Student Name:	VIVETHA B	Reg. No.:	192424170
Date:		Duration:	60 minutes

Code of Conduct

I VIVETHA B 192424170 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: VIVETHA B

Annexure A – Case Study Analysis

PHISHING EMAIL DETECTION AND CYBER THREAT INTELLIGENCE SYSTEM

1. Understand the Problem Statement

○ Explain the problem in your own words.

- Phishing emails are fraudulent messages created by attackers to trick users into sharing sensitive information such as passwords, banking details, and personal data. These emails may contain fake links, malicious attachments, or misleading content that appears to come from trusted sources.
- Identifying such emails manually can be difficult, especially when attackers use advanced techniques to make them look genuine.
- Traditional email security systems may also fail to detect newly developed phishing methods. The proposed **Phishing Email Detection and Cyber Threat Intelligence System** aims to automatically detect and classify suspicious emails.
- The system can analyse email content, sender details, URLs, attachments, and other suspicious patterns. Machine learning techniques can be used to identify whether an email is legitimate or phishing.

- Cyber Threat Intelligence can provide information about known malicious domains, IP addresses, and other cyber threats.
- By combining phishing detection with threat intelligence, the system can improve the accuracy of threat identification. It can generate alerts when a potentially dangerous email is detected and provide information about the detected threat.

- **Identify the objectives of the proposed system.**

The main objective of the proposed **Phishing Email Detection and Cyber Threat Intelligence System** is to develop an intelligent cybersecurity solution that can automatically detect phishing emails and provide useful information about potential cyber threats.

Specific Objectives

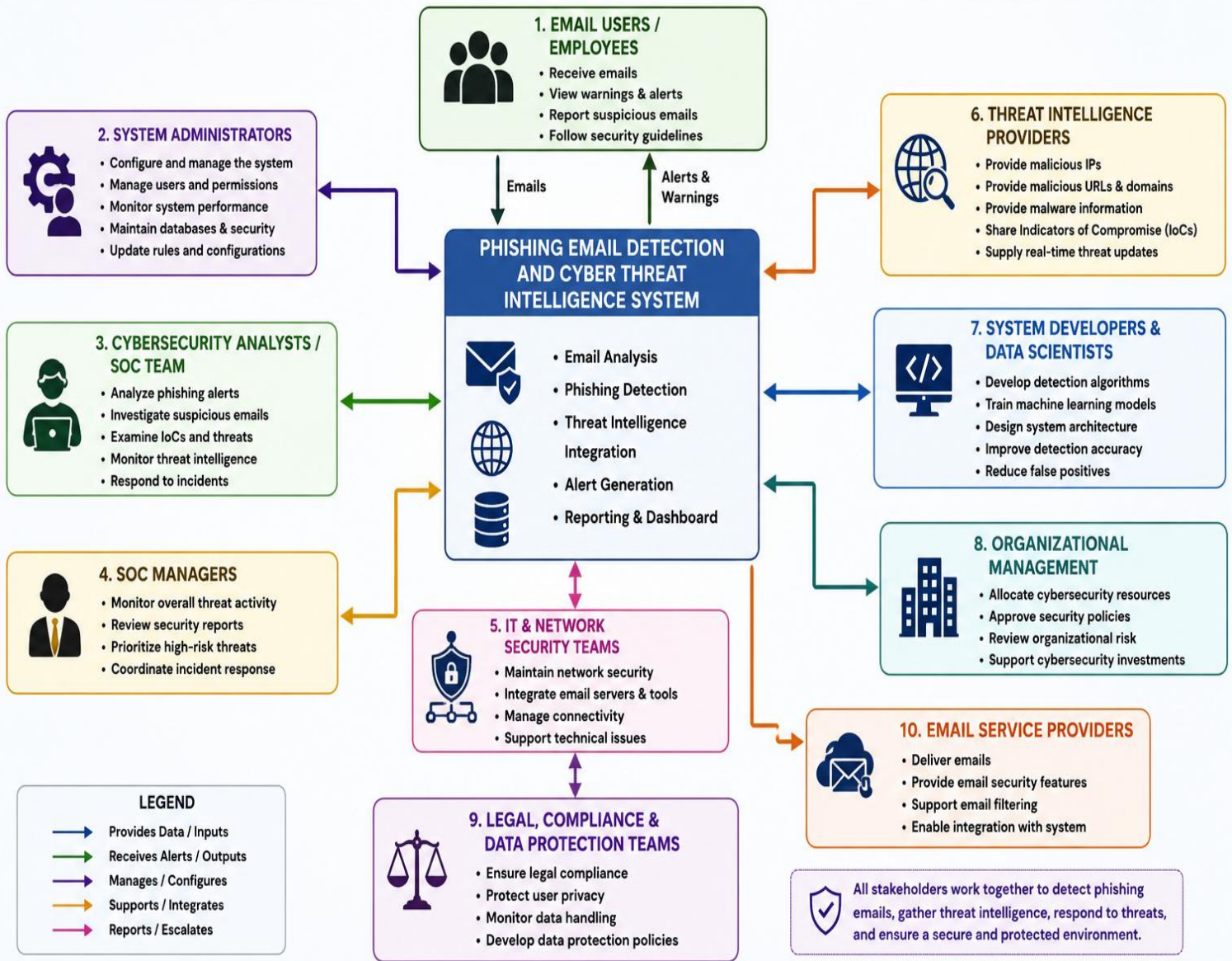
1. **To detect phishing emails automatically** by analysing email content, sender information, URLs, attachments, and header details.
2. **To classify emails accurately** as either legitimate or phishing using machine learning and intelligent detection techniques.
3. **To identify suspicious indicators** such as malicious URLs, IP addresses, domains, attachments, and other Indicators of Compromise (IoCs).
4. **To integrate Cyber Threat Intelligence** to collect, analyze, and correlate information about known and emerging cyber threats.
5. **To detect new and evolving phishing techniques** that may not be identified by traditional rule-based security systems.
6. **To generate real-time alerts and threat reports** when suspicious or malicious email activity is detected.
7. **To reduce false positives** by analyzing multiple email features before classifying an email as malicious.
8. **To provide explainable results** by showing why an email has been identified as phishing, such as suspicious links, unusual sender information, or malicious attachments.
9. **To improve organizational cybersecurity** by supporting faster threat detection, investigation, and response.
10. **To protect users from cyberattacks** by reducing the risk of credential theft, malware infection, financial fraud, and unauthorized access.

- **State the scope of the problem.**
- The scope of the proposed system focuses on detecting phishing emails and identifying related cyber threats.
- The system can analyze email messages to determine whether they are legitimate, suspicious, or phishing. It examines important features such as email content, sender details, URLs, and attachments.
- **It can reduce the time and effort required for manual phishing detection.** The system can be applied in organizations, educational institutions, businesses, and other environments where email security is important.
- The scope also includes improving the system using new datasets and updated threat intelligence. Overall, the proposed system aims to provide an effective and intelligent solution for phishing detection and cyber threat awareness.

2. Identify Key Stakeholders and Challenges

- **Identify all the stakeholders involved in the system.**
- The proposed Phishing Email Detection and Cyber Threat Intelligence System involves several important stakeholders. The primary stakeholders are the users who receive and interact with emails on a daily basis. The cybersecurity team is responsible for monitoring detected threats and responding to security incidents.
- System administrators manage the system, user accounts, configurations, and security settings. IT support teams help users handle phishing alerts and technical issues. Security analysts investigate suspicious emails and analyze cyber threat information provided by the system.
- Cyber Threat Intelligence providers supply information about malicious domains, IP addresses, URLs, and other threat indicators. Email service providers are involved in delivering and securing email communications. Government and cybersecurity regulatory authorities may provide security guidelines and compliance requirements.

STAKEHOLDER DIAGRAM – PHISHING EMAIL DETECTION AND CYBER THREAT INTELLIGENCE SYSTEM



○ Explain the roles and responsibilities of each stakeholder.

- The **users** are responsible for using the email system safely and reporting suspicious emails or phishing alerts. The **cybersecurity team** monitors security threats, investigates phishing incidents, and takes necessary actions to protect the organization.

- The **system administrators** manage user accounts, system configurations, access permissions, and overall system availability. The **IT support team** helps users resolve technical problems and provides assistance when phishing-related issues occur.
- **Security analysts** examine suspicious emails, URLs, domains, and other threat indicators to understand the nature of cyberattacks.
- The **organization management** provides resources, approves security policies, and ensures that the system supports organizational security goals.
- **Data scientists and machine learning engineers** develop, train, test, and improve machine learning models used for phishing email detection.
- **Software developers** design, implement, test, update, and maintain the application and its features. **Cyber Threat Intelligence providers** provide updated information about malicious IP addresses, domains, URLs, and other known cyber threats.
- **Email service providers** support secure email communication and help identify or block suspicious messages.
- **Government and cybersecurity authorities** provide security standards, regulations, and guidelines that help organizations maintain proper cybersecurity practices.
- **Database administrators** manage the storage, organization, backup, and security of email and threat-related data.
- Together, these stakeholders contribute to the effective detection of phishing emails and the protection of organizational information and systems.

- **Discuss the major challenges and issues faced by the stakeholders.**

- The stakeholders involved in the Phishing Email Detection and Cyber Threat Intelligence System face several challenges and security issues.
- Users may find it difficult to identify sophisticated phishing emails that look similar to genuine messages. Cybersecurity teams may receive a large number of alerts, making it difficult to investigate every threat quickly.
- System administrators may face challenges in maintaining system availability, access control, and security configurations. IT support teams may need to handle frequent user complaints and phishing-related incidents.
- Security analysts may struggle to analyze large amounts of email and threat intelligence data within a limited time. Organizations may face financial and operational risks if phishing attacks are not detected at an early stage.

- Data scientists may face difficulties in obtaining high-quality and updated datasets for training machine learning models. Software developers need to continuously update the system to handle new phishing techniques and cyber threats.
- Threat intelligence providers must maintain accurate and timely information about malicious domains, IP addresses, and URLs. Email service providers may face challenges in detecting new and constantly changing phishing patterns.
- Database administrators must protect sensitive email and threat data from unauthorized access. Government and cybersecurity authorities need to ensure that organizations follow changing security standards and regulations.
- Another major challenge is reducing false positives while maintaining high phishing detection accuracy. Overall, continuous monitoring, updated threat intelligence, secure data handling, and regular system improvements are necessary to overcome these challenges.

3. Analyze the Current Approach

- **Describe the existing system or current process.**
 - The existing process for detecting phishing emails mainly depends on traditional email security tools and manual user inspection. Email systems usually apply spam filters and rule-based security checks to identify suspicious messages.
 - These systems may examine the sender address, email content, links, attachments, and known malicious patterns. Emails identified as suspicious may be moved to a spam or junk folder. Users are also expected to check email addresses, links, and attachments before interacting with them. Organizations may use antivirus software, firewalls, and email security gateways to provide additional protection.
 - Security teams manually investigate suspicious emails reported by users. Cybersecurity teams may also compare suspicious URLs, IP addresses, and domains with available threat intelligence sources. Known malicious indicators can be blocked to prevent further attacks. However, attackers continuously develop new phishing techniques that may bypass existing security rules. Manual investigation can also take significant time when a large number of emails are received.
 - Traditional systems may produce false positives by incorrectly identifying legitimate emails as threats. They may also fail to provide detailed information about the reason an

email was classified as phishing. Therefore, the current approach has limitations in detecting advanced and constantly changing phishing attacks.

- **Identify its strengths and limitations.**

- The existing phishing email detection approach has several strengths that help protect users from common cyber threats. Traditional spam filters can quickly identify and block many unwanted and suspicious emails.
- Rule-based security systems can detect known phishing patterns, malicious links, and suspicious attachments. Antivirus and email security tools provide an additional layer of protection against harmful files and malware.

- Users can also manually verify suspicious emails before opening links or attachments. Security teams can investigate reported phishing emails and take corrective actions.

Existing threat intelligence sources can help identify previously reported malicious domains and IP addresses.

- However, the current approach also has several limitations. Rule-based systems may not effectively detect new and sophisticated phishing techniques.
- Attackers can modify email content, URLs, and sender information to bypass security rules. Manual email inspection requires time and cybersecurity knowledge from users. Security teams may become overloaded when handling a large number of suspicious emails.
- **Existing systems may generate false positives by incorrectly marking legitimate emails as phishing.** They may also generate false negatives by allowing sophisticated phishing emails to reach users.
- Traditional systems may not provide sufficient explanations about why an email was classified as malicious. Therefore, the current approach needs an intelligent and adaptive solution that can improve detection accuracy and respond effectively to emerging cyber threats.

- **Analyze the problems or gaps that need to be addressed.**

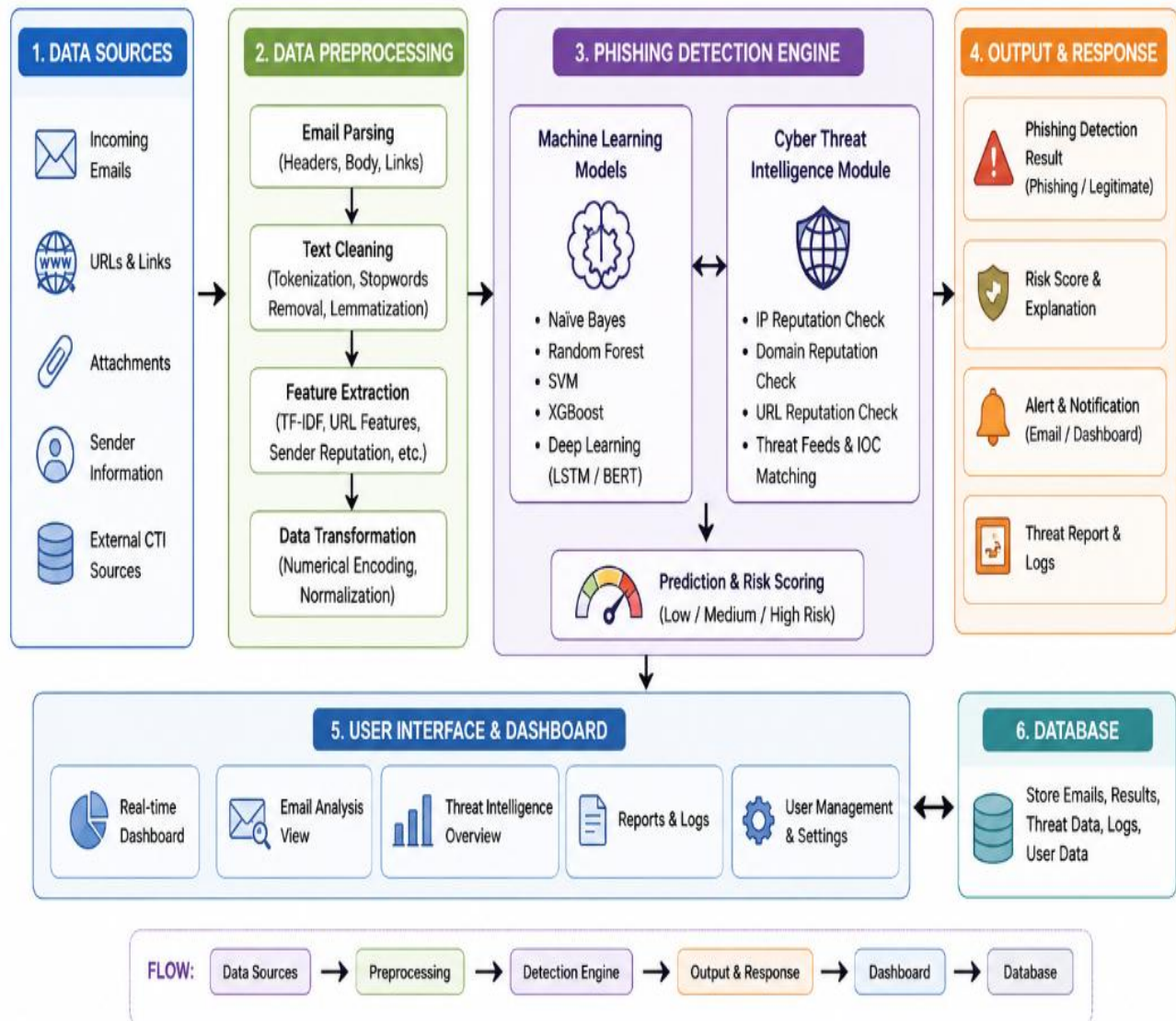
- The existing phishing email detection process has several problems and gaps that need to be addressed. Traditional rule-based systems mainly depend on predefined patterns and may fail to identify new and evolving phishing techniques.

- Attackers can easily modify email content, sender addresses, and URLs to bypass these security rules. Manual verification of emails is time-consuming and requires users to have sufficient cybersecurity awareness. Existing systems may produce false positives by marking genuine emails as suspicious, which can reduce user trust. They may also produce false negatives by failing to detect sophisticated phishing emails.
- Another major gap is the limited integration of real-time Cyber Threat Intelligence with email detection systems. Security teams may need to manually investigate suspicious URLs, domains, and IP addresses, which delays threat response. Existing systems may not provide clear explanations about why an email has been classified as phishing.
- They may also have difficulty handling large volumes of emails efficiently. Threat information may become outdated if the system does not regularly receive updated Threat Intelligence is needed to improve detection accuracy, reduce false results, provide useful threat information, and support faster security responses.

4. Propose a Suitable Solution Using Software Engineering Principles

- **Design a software-based solution for the given problem.**

PHISHING EMAIL DETECTION AND CYBER THREAT INTELLIGENCE SYSTEM – BLOCK DIAGRAM



The proposed solution is an intelligent **Phishing Email Detection and Cyber Threat Intelligence System** designed to automatically identify and analyze suspicious emails. The system will allow users to upload or forward emails for security analysis. First, the system will collect important email information such as sender details, subject, message content, URLs, and attachments. The collected data will then be preprocessed to remove unnecessary information and prepare it for analysis. A machine learning model will analyze the extracted features and classify the email as legitimate, suspicious, or phishing.

- **Describe the major functional modules of the proposed system.**

Module 1: Email Data Collection Module

This module collects email information for analysis.

Functions:

- Receive email data
- Extract sender and receiver information
- Collect email subject and body
- Extract URLs and attachments
- Collect email header information

Module 2: Email Preprocessing Module

This module prepares the email data for analysis.

Functions:

- Remove unnecessary characters
- Clean email text
- Normalize data
- Extract important keywords
- Process URLs and attachments

Module 3: Feature Extraction Module

This module identifies features that may indicate phishing activity.

Features include:

- Sender-domain mismatch
- Suspicious URLs
- Urgent or threatening language
- Unusual email structure
- Unknown attachments
- Suspicious domain names
- Email header anomalies

Module 4: Phishing Detection Module

This is the main intelligent detection component.

It uses:

- Machine learning models
- Rule-based detection
- Natural Language Processing
- Behavioural analysis

The system classifies an email as:

LEGITIMATE

|

|— SUSPICIOUS

|

|— PHISHING

Module 5: Cyber Threat Intelligence Module

This module collects and analyzes external threat information.

It examines:

- Malicious IP addresses
- Suspicious URLs
- Malicious domains
- Malware indicators
- Indicators of Compromise (IoCs)

The system correlates email indicators with known threat intelligence.

Module 6: Risk Assessment Module

This module assigns a risk score to each email.

Example:

Risk Score	Classification
0–30%	Low Risk
31–70%	Medium Risk
71–100%	High Risk

Module 7: Alert and Response Module

This module notifies users and security teams about detected threats.

Functions:

- Generate alerts
- Quarantine phishing emails
- Block malicious URLs
- Notify cybersecurity analysts
- Recommend response actions

Module 8: Dashboard and Reporting Module

This module provides a centralized interface for monitoring threats.

The dashboard can display:

- Total emails analyzed
- Number of phishing emails
- Risk levels
- Threat trends
- Malicious domains
- IP addresses
- Detection accuracy
- Recent alerts

Module 9: User and Access Management Module

This module controls access to the system.

Different users may have different permissions:

- Employee
- Analyst

- Administrator
- Manager
- Identify the functional and non-functional requirements.

Functional Requirements	Non-Functional Requirements
1. Email upload and submission	1. Accuracy
2. Email data extraction	2. Performance
3. Email preprocessing	3. Security
4. Phishing email classification	4. Usability
5. Suspicious URL detection	5. Reliability
6. Domain and IP analysis	6. Scalability
7. Cyber Threat Intelligence checking	7. Maintainability
8. Risk score calculation	8. Availability
9. Phishing alert generation	9. Compatibility
10. Threat explanation	10. Privacy
11. Threat data storage	11. Fast response time
12. Security monitoring dashboard	12. Data integrity
13. Threat detection reports	13. Regular updates

- **Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.**

➤ **Modularity** – The system is divided into separate modules such as email analysis, phishing detection, threat intelligence, alerts, and reporting.

- **Scalability** – The system can handle an increasing number of users, emails, and cyber threat records.
- **Maintainability** – The system is designed so that developers can easily update the software, detection models, and threat databases.
- **Reliability** – The system provides consistent phishing detection results and reduces system failures.
- **Usability** – A simple and user-friendly interface helps users easily upload emails and understand detection results.
- **Security** – User information, email content, and threat data are protected using authentication, authorization, and secure data handling.
- **Performance** – The system processes emails and generates phishing detection results within a short time.
- **Flexibility** – New phishing detection techniques, machine learning models, and threat intelligence sources can be added when required.
- **Testability** – Each module can be independently tested to identify errors and ensure that the complete system works correctly.
- **Data Integrity** – Email analysis results and threat intelligence data are stored accurately and protected from unauthorized modification.

4. Justify the Chosen Methodology/Framework

- **Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).**

The **Agile Software Development Model** is selected for developing the Phishing Email Detection and Cyber Threat Intelligence System. Agile is suitable because cybersecurity threats and phishing techniques change continuously. The system can be developed in small iterations called sprints. Each sprint can focus on developing a specific feature or module of the system. The first sprint can focus on email collection and preprocessing. The next sprint can implement phishing email detection using machine learning. Further sprints can add Cyber Threat Intelligence, threat scoring, alerts, and reporting features. Continuous testing can be performed during every development iteration. User feedback can be collected and used to improve the system in future sprints. New phishing patterns and threat intelligence can also be added whenever required. Agile allows

developers to identify and correct errors at an early stage. **It also supports changes in requirements without affecting the entire system. Regular communication among developers, security analysts, and users improves the quality of the solution.** The machine learning model can be continuously evaluated and improved using updated datasets. Security testing can also be performed regularly to protect sensitive email and threat information. Therefore, Agile is an appropriate methodology because it provides flexibility, continuous improvement, faster development, and better adaptation to changing cyber threats.

- **Justify why the selected methodology/framework is suitable for the proposed system.**

- Agile can adapt to continuously changing phishing techniques and cyber threats.
- The system can be developed step by step through different sprints.
- Each module can be tested regularly during development.
- New phishing patterns and threat intelligence can be added easily.
- Feedback from users and cybersecurity professionals can be included.
- Problems can be identified and corrected at an early stage.
- The machine learning model can be regularly improved using new datasets.
- Security testing can be performed throughout the development process.
- Development risks can be reduced by delivering features gradually.
- Requirements can be changed easily according to new cybersecurity needs.
- Agile improves communication between developers, analysts, and stakeholders.
- The system can be continuously improved after every iteration.
- Important features can be developed and delivered quickly.
- Threat detection performance can be regularly monitored and evaluated.
- Agile is suitable because it provides flexibility, continuous testing, regular updates, and quick response to emerging cyber threats.

- **Explain how it supports successful project development and maintenance.**

The Agile methodology supports successful development and maintenance of the Phishing Email Detection and Cyber Threat Intelligence System. Agile divides the project into small and

manageable development iterations. Each iteration focuses on developing and testing specific system features. This approach helps the development team complete important modules step by step. Continuous testing helps identify errors and problems at an early stage. Regular feedback from users and cybersecurity professionals helps improve the system. Agile allows new phishing techniques and cyber threats to be added quickly. The machine learning model can be updated regularly using new and relevant datasets. Security testing can be performed continuously to protect sensitive email information. The system can be modified easily when requirements or cybersecurity needs change. Regular monitoring helps maintain the performance and reliability of the system. Developers can fix bugs and improve existing features during maintenance iterations. Agile also encourages effective communication among developers, users, and security teams. Frequent updates help keep the system secure and useful over a long period. Therefore, Agile supports successful project development by providing flexibility, continuous improvement, testing, feedback, and effective long-term maintenance.

5. Discuss Expected Outcomes and Limitations

- **Describe the expected benefits of the proposed solution.**

The proposed Phishing Email Detection and Cyber Threat Intelligence System is expected to improve the overall security of email communication. The system can automatically analyze emails and identify phishing attempts without requiring users to manually inspect every message. It can examine email content, sender details, URLs, domains, IP addresses, and attachments to detect suspicious activities. Machine learning techniques can help improve the accuracy of phishing email classification. The integration of Cyber Threat Intelligence can provide updated information about known malicious domains, IP addresses, and other cyber threats. The system can generate alerts when a potentially dangerous email is detected. It can also provide a risk score and explanation to help users understand the nature of the threat. By detecting phishing attacks at an early stage, the system can reduce the risk of data theft, financial fraud, and unauthorized access. It can also reduce the workload of cybersecurity teams by automating the initial analysis process. Overall, the proposed solution can provide faster, intelligent, and reliable phishing detection for organizations.

- **Improved phishing detection** – Identifies suspicious and malicious emails effectively.
- **Faster threat response** – Provides alerts and threat information quickly.
- **Reduced manual effort** – Automates the initial email security analysis.

- **Better data protection** – Helps prevent data theft, fraud, and unauthorized access.
- **Enhanced cybersecurity** – Combines machine learning and Cyber Threat Intelligence

The system can therefore help organizations maintain better awareness of emerging phishing threats and take appropriate security measures. With regular updates, testing, and maintenance, the proposed solution can remain effective against changing cyber attack techniques and provide long-term protection for users and organizational data.

- **Explain how the solution addresses the identified challenges.**

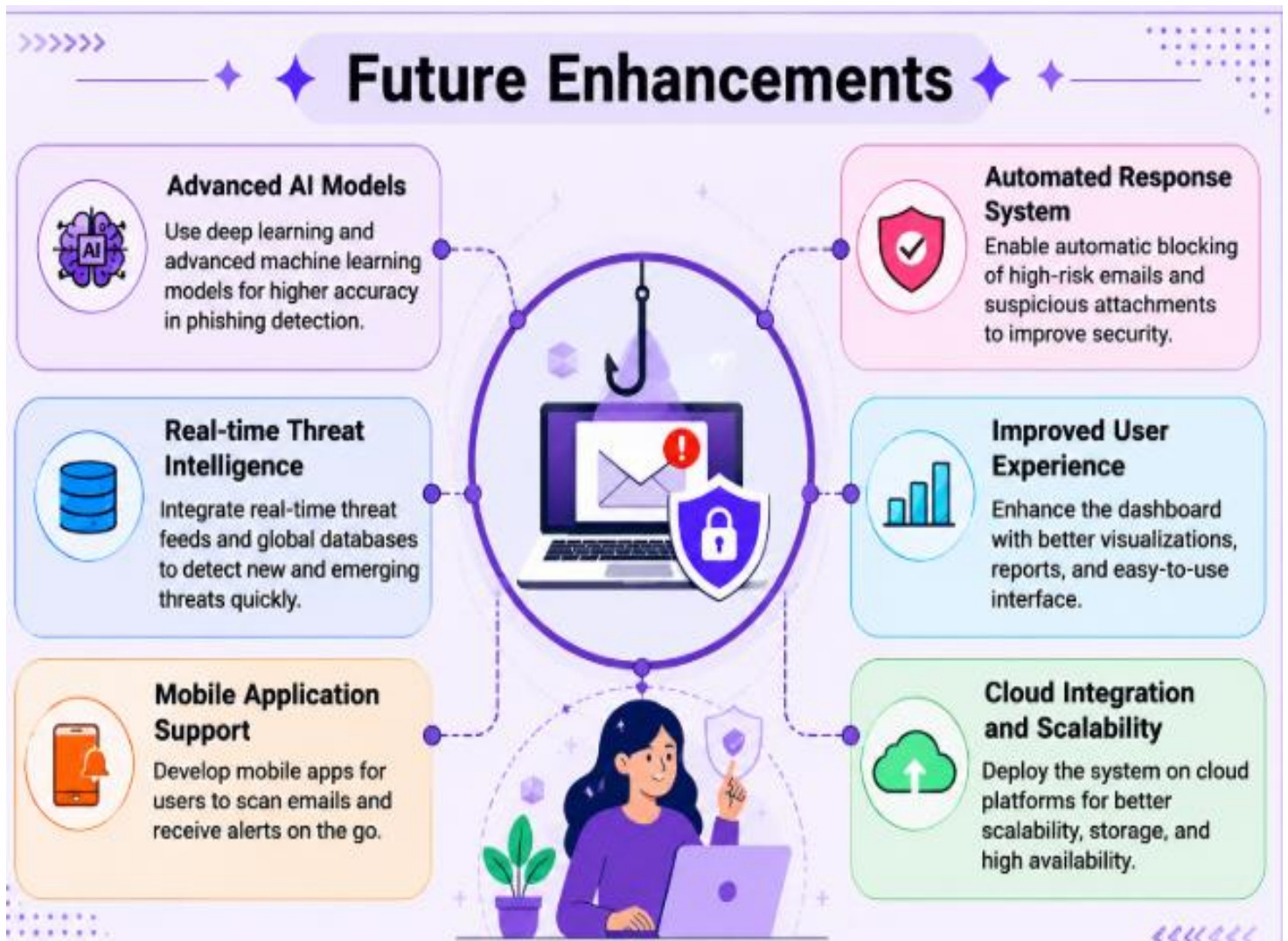
- **Detection of New and Sophisticated Phishing Attacks:** The system uses machine learning techniques to analyze email patterns, sender information, links, and other features, helping it identify new and sophisticated phishing attacks that traditional rule-based security systems may fail to detect.
- **Reduction of Manual Email Analysis:** The system automatically examines email content, URLs, domains, IP addresses, and attachments, reducing the need for users and security teams to manually inspect every email and saving significant time and effort.
- **Updated Cyber Threat Intelligence:** The system integrates Cyber Threat Intelligence to obtain updated information about malicious URLs, domains, IP addresses, and other indicators of compromise, helping the system identify known and emerging cyber threats.
- **Improved Detection Accuracy:** The machine learning model can be continuously trained and updated using new phishing email datasets, which helps reduce false positives and false negatives while improving the overall accuracy and reliability of phishing detection.
- **Faster Threat Response and Better Understanding:** The system provides real-time alerts, risk scores, and explanations when suspicious emails are detected, allowing users and cybersecurity teams to quickly understand the threat, take appropriate action, and reduce potential security damage.

- **Discuss potential risks, implementation challenges, and limitations.**

- The proposed Phishing Email Detection and Cyber Threat Intelligence System may face several risks and challenges during implementation.
- Machine learning models may produce false positives by classifying genuine emails as phishing or false negatives by allowing dangerous emails to pass.

- The system requires a large and high-quality dataset containing updated phishing and legitimate emails for effective training.
- Cyber attackers continuously develop new techniques, making it difficult to keep the detection model updated. Threat intelligence sources may sometimes contain incomplete, outdated, or inaccurate information.
- Handling sensitive email content also creates privacy and data security concerns. The system may require significant computing resources when analyzing a large number of emails.
- Integration with existing email platforms and security tools may also be technically challenging. Regular maintenance and model retraining are required to maintain detection accuracy.
- Users may also ignore security alerts if the system generates too many notifications. Therefore, continuous monitoring, security testing, regular updates, and proper data protection are necessary to overcome these limitations.

- **Suggest possible future enhancements.**



The proposed system can be enhanced in the future by using advanced AI and deep learning models for more accurate phishing detection. Real-time Cyber Threat Intelligence can be integrated to identify newly emerging threats quickly. Automated response features can block or quarantine dangerous emails and suspicious links. A mobile application can allow users to scan emails and receive security alerts anytime. Cloud integration and improved user experience can make the system more scalable, accessible, and efficient.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.

Methodology/ Framework Justification	Selects the most suitable SDLC/model/frame work and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.
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Criteria	Mark
Problem Understanding	/5
Stakeholder & Challenge Analysis	/5
Current System Analysis	/5
Proposed Software Solution & Software Engineering Principles	/5
Methodology/ Framework Justification	/5
TOTAL	/5